



Lists & Tuples

Topics:

- Understanding List & Tuples
- Associated Methods
- Indexing and Slicing

Lists

List

Lists are just like the arrays, declared in other languages which is a ordered collection of data. It is very flexible as the items in a list do not need to be of the same type.

Creating List

Lists in Python can be created by just placing the sequence inside the square brackets[].

Examples

```
>>> x = [10,20,30,40,50,60]
```

```
>>> print(x)
```

```
[10, 20, 30, 40, 50, 60]
```

```
>>> print(type(x))
```

```
<class 'list'>
```

List Methods

List Object is defined with the following pre-defined Methods

`.append(object)`

Appends the given object to the end of the List

`.clear()`

It clears the Elements of the List(It makes the list Empty)

`.copy()`

It creates a shallow copy of the List object

List Methods

`.count(value)`

Returns the count of the occurrence of the given value in the list

`.index(value, start, end)`

returns the index of the given value in the list in between the start and end position(start is included and end is excluded)

`.insert(index, value)`

It inserts the value before the given index in the list

List Methods

`.pop(index)`

Removes the element at the given index(default is -1) and returns the removed element

`.remove(value)`

removes the first occurrence of the given value from the list

`.reverse()`

Reverses the elements in the list and assigned back to list

List Methods

`.sort(reverse=True)`

Sorts the elements in the list either in ascending or descending order and assigned back to list.

List Examples

```
>>> x = [10,20,30,40,50]
```

```
>>> print(x)
```

```
[10, 20, 30, 40, 50]
```

```
>>> x.append(60)
```

```
>>> print(x)
```

```
[10, 20, 30, 40, 50, 60]
```

```
>>> x.clear()
```

```
>>> print(x)
```

```
[]
```


List Examples

```
>>> x = [10,20,30,40,50]
```

```
>>> print(x)
```

```
[10, 20, 30, 40, 50]
```

```
>>> y = x.copy()
```

```
>>> id(x)
```

```
2133628180608
```

```
>>> id(y)
```

```
2133628115136
```

List Examples

```
>>> x = [10,20,30,10,20,10,50]
```

```
>>> print(x)
```

```
[10, 20, 30, 10, 20, 10, 50]
```

```
>>> print(x.count(10))
```

```
3
```

```
>>> print(x.count(20))
```

```
2
```

List Examples

```
>>> print(x.index(10,3,6))
```

```
3
```

```
>>> x.insert(2,100)
```

```
>>> print(x)
```

```
[10, 20, 100, 30, 10, 20, 10, 50]
```

```
>>> x.pop(2)
```

```
100
```

```
>>> print(x)
```

```
[10, 20, 30, 10, 20, 10, 50]
```

```
>>> x.pop()
```

```
50
```

```
>>> print(x)
```

```
[10, 20, 30, 10, 20, 10]
```

List Examples

```
>>> x=[10,40,20,50,60,30]
```

```
>>> x.reverse()
```

```
>>> print(x)
```

```
[30, 60, 50, 20, 40, 10]
```

```
>>> x.sort(reverse=True)
```

```
>>> print(x)
```

```
[60, 50, 40, 30, 20, 10]
```

```
>>> x.sort(reverse=False)
```

```
>>> print(x)
```

```
[10, 20, 30, 40, 50, 60]
```

Tuples

Tuple

Tuples are just same as list but immutable Objects i.e., they cannot be modified.

Creating Tuple

Tuples in Python can be created by just placing the sequence inside the parenthesis()

Examples

```
>>> x = (10,20,30,40,50,60)
```

```
>>> print(x)
```

```
(10, 20, 30, 40, 50, 60)
```

```
>>> print(type(x))
```

```
<class 'tuple'>
```

Tuple Methods

Tuple Object is defined with the following pre-defined Methods

`.count(value)`

Returns the count of the value in the Tuple

`.index(value, stop, end)`

Returns the index of the given value between the stop and end position.

Examples

```
>>> x = (10,20,30,10,30,20)
```

```
>>> x.count(10)
```

```
2
```

```
x.index(10)
```

```
0
```

```
>>>x.index(10,3,5)
```

```
3
```


Indexing and Slicing

Indexing

- It is a process of extracting an element from a list or Tuple.

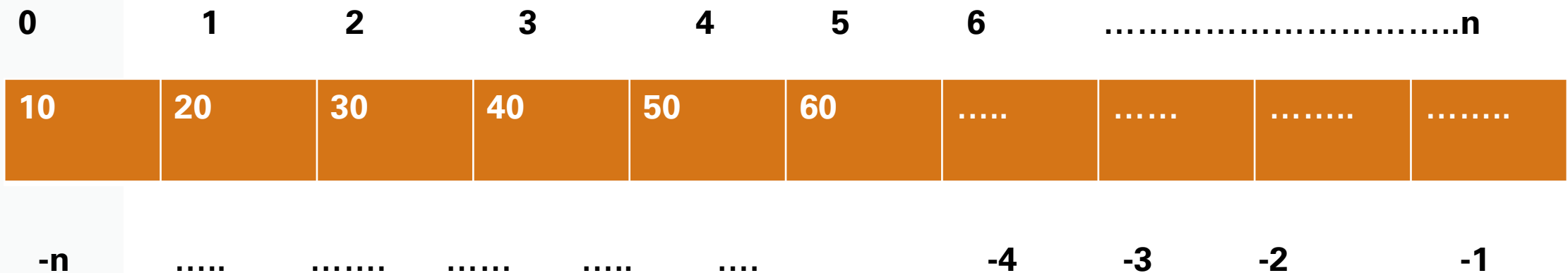
Slicing

- It is a process of Subsetting a set of elements from a list or tuple.

Indexes in list or tuple

List[index] → Indexing

List[start : end : step] → Slicing



Note: In Python Index starts from 0 to n from the left to right hand side and from -1 to -n from the Right to Left hand side.

Examples

```
>>> x = [10,20,30,40,50,60,70,80]
```

```
>>> x[4]
```

```
50
```

```
>>> x[-1]
```

```
80
```

```
>>> x[2:6]
```

```
[30, 40, 50, 60]
```

```
>>> x[1:-1]
```

```
[20, 30, 40, 50, 60, 70]
```

```
>>> x[:5]
```

```
[10, 20, 30, 40, 50]
```

```
>>> x[2:]
```

```
[30, 40, 50, 60, 70, 80]
```

Examples

```
>>> x[::2]
```

```
[10, 30, 50, 70]
```

```
>>> x[::-1]
```

```
[80, 70, 60, 50, 40, 30, 20, 10]
```

```
>>> x[2::3]
```

```
[30, 60]
```

```
>>> x[-2::-2]
```

```
[70, 50, 30, 10]
```